MA1522 Linear Algebra for Computing
L2 Lesson 19



Tan Kai Meng

Semester 1, 2026/27

Linear Transformations

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if and only if $T(c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2) = c_1T(\boldsymbol{v}_1) + c_2T(\boldsymbol{v}_2)$ for all $c_1, c_2 \in \mathbb{R}$ and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^n$.

Remarks

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a function.

1. T is a linear transformation if and only if $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^n$, and $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$.
2. If T is a linear transformation, then $T(\mathbf{0}_n) = \mathbf{0}_m$.

Linear Transformations

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if and only if $T(c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2) = c_1T(\boldsymbol{v}_1) + c_2T(\boldsymbol{v}_2)$ for all $c_1, c_2 \in \mathbb{R}$ and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^n$.

Remarks

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a function.

1. T is a linear transformation if and only if $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^n$, and $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$.
2. If T is a linear transformation, then $T(\mathbf{0}_n) = \mathbf{0}_m$.

Linear Transformations

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if and only if $T(c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2) = c_1T(\boldsymbol{v}_1) + c_2T(\boldsymbol{v}_2)$ for all $c_1, c_2 \in \mathbb{R}$ and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^n$.

Remarks

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a function.

1. T is a linear transformation if and only if $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^n$, and $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$.
2. If T is a linear transformation, then $T(\mathbf{0}_n) = \mathbf{0}_m$.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$.

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$.

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$.

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$.

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$.

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$.

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$.

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$.

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$.

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$.

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$.

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$. ..... **Ans: False**

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$.

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$.

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$.

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$.

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$. ..... **Ans: False**

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$. ..... **Ans: True**

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$.

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$.

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$.

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$. ..... **Ans: False**

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$. ..... **Ans: True**

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$. ..... **Ans: False**

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$.

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$.

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$. ..... **Ans: False**

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$. ..... **Ans: True**

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$. ..... **Ans: False**

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$. ..... **Ans: True**

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$.

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$. ..... **Ans: False**

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$. ..... **Ans: True**

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$. ..... **Ans: False**

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$. ..... **Ans: True**

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$. ..... **Ans: False**

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.

Question 1

PollEv.com/kaimengtan612

Which of the following functions are linear transformations:

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$. ..... **Ans: True**

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ 0 \end{pmatrix}$. ..... **Ans: False**

c. $T_3 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$. ..... **Ans: True**

d. $T_4 : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$. ..... **Ans: False**

e. $T_5 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$. ..... **Ans: True**

f. $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{x}$. ..... **Ans: False**

g. $T_7 : \mathbb{R}^n \to \mathbb{R}^m$ such that $T_7(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ is a fixed $m \times n$ matrix.
..... **Ans: True**

Question 2

PollEv.com/kaimengtan612

Construct a function $T : \mathbb{R}^2 \to \mathbb{R}$ such that $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^2$, but $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) \neq T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for some $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^2$.

Question 2

PollEv.com/kaimengtan612

Construct a function $T : \mathbb{R}^2 \to \mathbb{R}$ such that $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^2$, but $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) \neq T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for some $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^2$.

Solution.

Let $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} x, & \text{if } y = 0; \\ 0, & \text{if } y \neq 0. \end{cases}$

Question 2

PollEv.com/kaimengtan612

Construct a function $T : \mathbb{R}^2 \to \mathbb{R}$ such that $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^2$, but $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) \neq T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for some $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^2$.

Solution.

Let $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} x, & \text{if } y = 0; \\ 0, & \text{if } y \neq 0. \end{cases}$

Then $T\left(c\begin{pmatrix} x \\ y \end{pmatrix}\right) = T\begin{pmatrix} cx \\ cy \end{pmatrix} = \begin{cases} cx, & \text{if } cy = 0; \\ 0, & \text{if } cy \neq 0 \end{cases} = \begin{cases} cx, & \text{if } y = 0; \\ 0, & \text{if } y \neq 0 \end{cases} = cT\begin{pmatrix} x \\ y \end{pmatrix}.$

Question 2

PollEv.com/kaimengtan612

Construct a function $T : \mathbb{R}^2 \to \mathbb{R}$ such that $T(c\boldsymbol{v}) = cT(\boldsymbol{v})$ for all $c \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^2$, but $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) \neq T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for some $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^2$.

Solution.

Let $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} x, & \text{if } y = 0; \\ 0, & \text{if } y \neq 0. \end{cases}$

Then $T\left(c\begin{pmatrix} x \\ y \end{pmatrix}\right) = T\begin{pmatrix} cx \\ cy \end{pmatrix} = \begin{cases} cx, & \text{if } cy = 0; \\ 0, & \text{if } cy \neq 0 \end{cases} = \begin{cases} cx, & \text{if } y = 0; \\ 0, & \text{if } y \neq 0 \end{cases} = cT\begin{pmatrix} x \\ y \end{pmatrix}.$

Let $\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\boldsymbol{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Then $T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2) = 0 + 0 = 0$, but $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T\begin{pmatrix} 2 \\ 0 \end{pmatrix} = 2$. □

Standard Matrix of Linear Transformation

Theorem

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique $m \times n$ matrix $\boldsymbol{A}$ such that $T(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$ for all $\boldsymbol{x} \in \mathbb{R}^n$, namely

$$\boldsymbol{A} = \begin{pmatrix} T\begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} & T\begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} & \cdots & T\begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \end{pmatrix}.$$

We call the $m \times n$ matrix $\boldsymbol{A}$ is the **standard matrix** *of T, denoted as $[T]_E$, or simply $[T]$. Here E denotes the standard basis for $\mathbb{R}^n$.*

Question 3

PollEv.com/kaimengtan612

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ and $T' : \mathbb{R}^m \to \mathbb{R}^p$ be linear transformations. Show that $T' \circ T : \mathbb{R}^n \to \mathbb{R}^p$ is a linear transformation, and

$$[T' \circ T] = [T'][T].$$

(Hint: $(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x}))$ for all $\boldsymbol{x} \in \mathbb{R}^n$.)

Question 3

PollEv.com/kaimengtan612

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ and $T' : \mathbb{R}^m \to \mathbb{R}^p$ be linear transformations.
Show that $T' \circ T : \mathbb{R}^n \to \mathbb{R}^p$ is a linear transformation, and

$$[T' \circ T] = [T'][T].$$

(Hint: $(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x}))$ for all $\boldsymbol{x} \in \mathbb{R}^n$.)

Solution.

$$(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x})) = [T'](T(\boldsymbol{x})) = [T']([T]\boldsymbol{x}) = ([T'][T])\boldsymbol{x}.$$

Question 3

PollEv.com/kaimengtan612

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ and $T' : \mathbb{R}^m \to \mathbb{R}^p$ be linear transformations. Show that $T' \circ T : \mathbb{R}^n \to \mathbb{R}^p$ is a linear transformation, and

$$[T' \circ T] = [T'][T].$$

(Hint: $(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x}))$ for all $\boldsymbol{x} \in \mathbb{R}^n$.)

Solution.

$$(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x})) = [T'](T(\boldsymbol{x})) = [T']([T]\boldsymbol{x}) = ([T'][T])\boldsymbol{x}.$$

Since $[T'][T]$ is a $p \times n$ matrix, $T' \circ T$ is a linear transformation (by Question 1(g)).

Question 3

PollEv.com/kaimengtan612

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ and $T' : \mathbb{R}^m \to \mathbb{R}^p$ be linear transformations. Show that $T' \circ T : \mathbb{R}^n \to \mathbb{R}^p$ is a linear transformation, and

$$[T' \circ T] = [T'][T].$$

(Hint: $(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x}))$ for all $\boldsymbol{x} \in \mathbb{R}^n$.)

Solution.

$$(T' \circ T)(\boldsymbol{x}) = T'(T(\boldsymbol{x})) = [T'](T(\boldsymbol{x})) = [T']([T]\boldsymbol{x}) = ([T'][T])\boldsymbol{x}.$$

Since $[T'][T]$ is a $p \times n$ matrix, $T' \circ T$ is a linear transformation (by Question 1(g)).

Furthermore the equation also shows that the standard matrix of $T' \circ T$ is equal to $[T'][T]$. □

Change of Basis

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n\}$ be a basis for $\mathbb{R}^n$.

For $\boldsymbol{v} \in \mathbb{R}^n$, we have $\boldsymbol{v} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n$ for some unique $c_1, c_2, \ldots, c_n \in \mathbb{R}^n$. Thus

$$\begin{aligned} T(\boldsymbol{v}) &= T(c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n) \\ &= c_1T(\boldsymbol{u}_1) + c_2T(\boldsymbol{u}_2) + \cdots + c_nT(\boldsymbol{u}_n) \\ &= \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = [T]_S[\boldsymbol{v}]_S, \end{aligned}$$

where

$$[T]_S = \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix}$$

is the **matrix representing T with respect to S**.

Note

The standard matrix $[T]$ of T is equal to $[T]_E$ where E is the standard basis for $\mathbb{R}^n$, so $[T]_S$ generalises the notion of matrix representing T.

Change of Basis

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n\}$ be a basis for $\mathbb{R}^n$.

For $\boldsymbol{v} \in \mathbb{R}^n$, we have $\boldsymbol{v} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n$ for some unique $c_1, c_2, \ldots, c_n \in \mathbb{R}^n$. Thus

$$\begin{aligned} T(\boldsymbol{v}) &= T(c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n) \\ &= c_1T(\boldsymbol{u}_1) + c_2T(\boldsymbol{u}_2) + \cdots + c_nT(\boldsymbol{u}_n) \\ &= \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = [T]_S[\boldsymbol{v}]_S, \end{aligned}$$

where

$$[T]_S = \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix}$$

is the **matrix representing T with respect to S**.

Note

The standard matrix $[T]$ of T is equal to $[T]_E$ where E is the standard basis for $\mathbb{R}^n$, so $[T]_S$ generalises the notion of matrix representing T.

Change of Basis

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n\}$ be a basis for $\mathbb{R}^n$.

For $\boldsymbol{v} \in \mathbb{R}^n$, we have $\boldsymbol{v} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n$ for some unique $c_1, c_2, \ldots, c_n \in \mathbb{R}^n$. Thus

$$\begin{aligned} T(\boldsymbol{v}) &= T(c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n) \\ &= c_1T(\boldsymbol{u}_1) + c_2T(\boldsymbol{u}_2) + \cdots + c_nT(\boldsymbol{u}_n) \\ &= \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = [T]_S[\boldsymbol{v}]_S, \end{aligned}$$

where

$$[T]_S = \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix}$$

is the **matrix representing T with respect to S**.

Note

The standard matrix $[T]$ of T is equal to $[T]_E$ where E is the standard basis for $\mathbb{R}^n$, so $[T]_S$ generalises the notion of matrix representing T.

Relation with Transition Matrices

$$\begin{aligned}
[T]_S &= \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix} \\
&= \begin{pmatrix} [T]\boldsymbol{u}_1 & [T]\boldsymbol{u}_2 & \cdots & [T]\boldsymbol{u}_n \end{pmatrix} \\
&= [T] \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \end{pmatrix} \\
&= [T] \begin{pmatrix} [\boldsymbol{u}_1]_E & [\boldsymbol{u}_2]_E & \cdots & [\boldsymbol{u}_n]_E \end{pmatrix}.
\end{aligned}$$

Note that $\begin{pmatrix} [\boldsymbol{u}_1]_E & [\boldsymbol{u}_2]_E & \cdots & [\boldsymbol{u}_n]_E \end{pmatrix}$ is the transition matrix from S to E. Thus,

$$[T]_S = [T] \times \text{transition matrix from } S \text{ to } E;$$
$$[T] = [T]_S \times \text{transition matrix from } E \text{ to } S.$$

Relation with Transition Matrices

$$\begin{aligned}
[T]_S &= \begin{pmatrix} T(\boldsymbol{u}_1) & T(\boldsymbol{u}_2) & \cdots & T(\boldsymbol{u}_n) \end{pmatrix} \\
&= \begin{pmatrix} [T]\boldsymbol{u}_1 & [T]\boldsymbol{u}_2 & \cdots & [T]\boldsymbol{u}_n \end{pmatrix} \\
&= [T] \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \end{pmatrix} \\
&= [T] \begin{pmatrix} [\boldsymbol{u}_1]_E & [\boldsymbol{u}_2]_E & \cdots & [\boldsymbol{u}_n]_E \end{pmatrix}.
\end{aligned}$$

Note that $\begin{pmatrix} [\boldsymbol{u}_1]_E & [\boldsymbol{u}_2]_E & \cdots & [\boldsymbol{u}_n]_E \end{pmatrix}$ is the transition matrix from S to E. Thus,

$$[T]_S = [T] \times \text{transition matrix from } S \text{ to } E;$$
$$[T] = [T]_S \times \text{transition matrix from } E \text{ to } S.$$

Question 4

PollEv.com/kaimengtan612

For this question, you may assume that

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

is a basis for $\mathbb{R}^3$.
Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation such that

$$T \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad T \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Find $T(\boldsymbol{v})$, where $\boldsymbol{v} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$.

Question 4 — Solution (Method 1)

$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S$:

We have $[T]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

Find $[\boldsymbol{v}]_S$ by solving

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} [\boldsymbol{v}]_S = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \Rightarrow \quad [\boldsymbol{v}]_S = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Question 4 — Solution (Method 1)

$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S$:

We have $[T]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

Find $[\boldsymbol{v}]_S$ by solving

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} [\boldsymbol{v}]_S = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \Rightarrow \quad [\boldsymbol{v}]_S = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Question 4 — Solution (Method 1)

$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S$:

We have $[T]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

Find $[\boldsymbol{v}]_S$ by solving

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} [\boldsymbol{v}]_S = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \Rightarrow \quad [\boldsymbol{v}]_S = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Question 4 — Solution (Method 1)

$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S$:

We have $[T]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

Find $[\boldsymbol{v}]_S$ by solving

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} [\boldsymbol{v}]_S = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \Rightarrow \quad [\boldsymbol{v}]_S = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = [T]_S[\boldsymbol{v}]_S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Question 4 — Solution (Method 2)

$T(\boldsymbol{v}) = [T]\boldsymbol{v}$:

We have $[T]_S = [T] \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$$\Rightarrow \quad [T] = [T]_S \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Question 4 — Solution (Method 2)

$T(\boldsymbol{v}) = [T]\boldsymbol{v}$:

We have $[T]_S = [T] \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$$\Rightarrow \quad [T] = [T]_S \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Question 4 — Solution (Method 2)

$T(\boldsymbol{v}) = [T]\boldsymbol{v}$:

We have $[T]_S = [T] \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$$\Rightarrow \quad [T] = [T]_S \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

Thus

$$T(\boldsymbol{v}) = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 6 \end{pmatrix}.$$

Next Lesson ...

We will explore **Sections 7.1 and 7.2**.
Please watch/review the video clips for these sections before then.